

CourtMate Technical Documentation

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1. Project Description

CourtMate is an AI-assisted racket-sports platform that connects **game discovery, session coordination, and player development** in one experience. It supports pickleball, badminton, tennis, padel, squash, and table tennis through a responsive web application.

Organizing recreational sports often requires disconnected messaging groups, manual player searches, and informal skill assessments. CourtMate brings these activities into a structured workflow: players discover suitable sessions, organizers manage participation and replacements, and completed games contribute to a persistent record of activity and skill.

The **Ask CourtMate assistant** is a primary interaction layer. Players can describe what they want in everyday language, refine a search through conversation, or create a game through guided questions. The backend validates the resulting request against application rules and stored records before returning results or executing an action.

CourtMate also maintains a **sport-specific CourtMate Rating (CMR), ranging from 1.00 to 10.00**. CMR provides a common skill reference for matchmaking and progression. Where competitive result processing is enabled, confirmed results can adjust ratings using expected team strength, match outcome, score margin, and rating confidence. Skill ratings and participation reliability serve distinct purposes: one supports balanced games, while the other helps organizers assess attendance risk.

The intended impact is practical: **reduce coordination effort, improve game compatibility, fill vacant spots, and help players participate consistently.**

2. Project Use Case

Discover → Organize → Join or Waitlist → Coordinate → Play → Record Feedback and Results → Build Reputation → Play Again

1. **Authenticate and establish a profile**

Players sign in with Google through Firebase Authentication. The frontend attaches a Firebase ID token to authenticated API requests, and FastAPI verifies identity before accessing protected resources.

2. **Express intent and discover games**

Players browse Explore and the community map or ask a question such as, “Find a

pickleball game near Whitefield tomorrow evening.” The assistant extracts search criteria and requests missing details when needed.

3. **Evaluate eligibility and compatibility**

The backend checks sport, date, time, location, travel distance, visibility, capacity, and skill compatibility. Semantic retrieval can identify candidates, but server-side rules determine which sessions are eligible to appear.

4. **Create a session or request a place**

Organizers define the sport, area, schedule, format, player capacity, and visibility. Players request access to existing sessions, with approval and waitlist behavior governed by the session’s rules.

5. **Manage participation and replacements**

CourtMate tracks confirmed players, pending requests, withdrawals, and ordered waitlists. When a place becomes available, waitlist promotion and substitute recommendations help organizers restore the lineup.

6. **Coordinate through the Rally Circle**

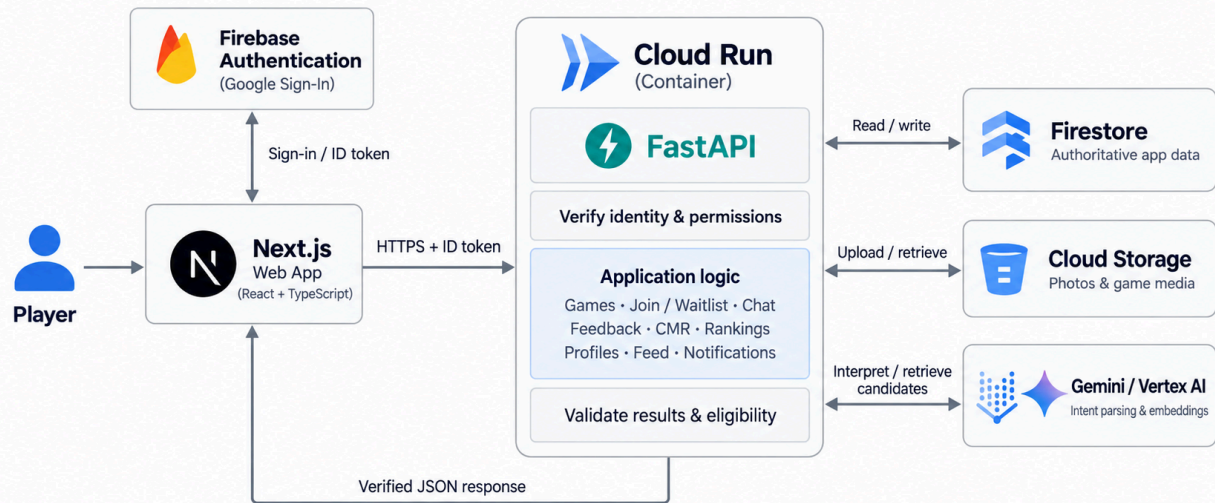
Authorized participants access the session workspace for game details, the player lineup, chat, and relevant participation information. Access is checked on the server against the player’s relationship to the session.

7. **Capture feedback and build progression**

After completion, players submit feedback and, where supported, record competitive results. Sport-specific CMR, activity history, and reliability information support future matchmaking, while profiles, follows, and social posts help players stay connected.

3. Architecture Diagram

CourtMate Architecture



AI interprets requests. FastAPI validates results against stored records.

FastAPI enforces authorization and business rules, with Firestore as the source of truth. Gemini and Vertex AI assist intent parsing and retrieval; all results are validated by the backend before reaching the user.